# Spatial Analysis

May 16, 2023

## 1 Methods

For each sample, we will compare the distribution of each markers in the normal and lesion regions.

- Markers for DD: CD3, CD4, CD8a, CD11c, CD15, CD45, CD68, CD69, CD123, CD279
- Markers for CF: CD1a, CD3, CD4, CD8a, CD11c, CD15, CD45, CD69, CD123, CD279 (CD68 did not pass QC)

For CD123, segmentation is done at flow=0.5 to get more masks in the bright area and a threshold of 5000 is set for total intensity to remove masks in the debris field in the lesion area.

The size of the skin tissue lesion area is determined by morphology using wheat germ agglutinin (WGA) flourescent stain. ImageJ is used to demarcate the lesion area, then a straight line ROI is drawn to span from one side of the lesion to the other side of the lesion, with the endpoints of the line positioned at the outermost skin boundary.

Some considerations on region boundary and distance metric:
1. Should we draw the boundary between normal and lesion based on WGA only or all markers?
2. Shall each ROI be convex?

For (1), the two choices lead are answering different questions. Drawing boundary based on WGA only answers a stronger question than drawing the boundary based on all markers.

For (2), a convex shape is a shape that a line connecting any two points in the shape completely lies within the shape. Practically speaking, in this context it is equivalent to drawing a straight line to separate normal from lesion. Pros for straight boundary: the span between any cell and the closest point on the skin lies completely within the region. So the distance measure is well defined. In contrast, if a line from a cell to its closest point on the skin surface cuts through both normal and lesion regions, the interpretation of distance to skin surface is muddled. Pros for curved boundary: Regions (which is a type of neighborhoods) do not necessarily have geometric shapes.

These two questions are also intertwined. If we use WGA only to draw boundaries, we can only use straight line boundaries because the difference between normal and lesion is only visible in epidermis in WGA staining. A straight line can be drawn in epidermis and extend to dermis, but not a curve.

## 2 Results

| | Counts | | | | Counts/mm2 | | | |
|---|---|---|---|---|---|---|---|---|
| | normal | | lesion | | normal | | lesion | |
| | epi | dermis | epi | dermis | epi | dermis | epi | dermis |
| CD3 | 74 | 1222 | 158 | 5395 | 631 | 2874 | 121 | 1399 |
| CD15 | 11 | 238 | 6567 | 2867 | 93 | 559 | 5050 | 743 |
| CD123 | 131 | 879 | 1350 | 8038 | 1117 | 2067 | 1038 | 2084 |
| CD11c | 343 | 2056 | 8029 | 17905 | 2926 | 4836 | 6174 | 4643 |
| CD68 | 154 | 998 | 2524 | 11081 | 1314 | 2347 | 1941 | 2874 |
| area | 0.12 | 0.43 | 1.30 | 3.86 | | | | |

Table 1: DD

| | Counts | | | | Counts/mm2 | | | |
|---|---|---|---|---|---|---|---|---|
| | normal | | lesion | | normal | | lesion | |
| | epi | dermis | epi | dermis | epi | dermis | epi | dermis |
| CD3 | 422 | 8435 | 331 | 5629 | 976 | 4476 | 819 | 5574 |
| CD15 | 7 | 2154 | 756 | 391 | 16 | 1143 | 1871 | 387 |
| CD123 | 76 | 3680 | 324 | 1298 | 175 | 1952 | 801 | 1285 |
| CD11c | 624 | 8498 | 1308 | 4299 | 1443 | 4509 | 3237 | 4257 |
| area | 0.43 | 1.88 | 0.40 | 1.01 | | | | |

Table 2: CF

| | DD | CF | DD/CF |
|---|---|---|---|
| lesion width | 2862 | 1202 | 2.38 |
| normal width | 251 | 1531 | 0.16 |
| separation between CD3/CD15 peaks (ref: skin) | 607 | 427 | 1.42 |
| separation between CD3/CD15 peaks (ref: DEJ) | 319 | 337 | 0.95 |
| CD3 epi | 121 | 819 | 0.15 |
| CD3 dermis | 1399 | 5574 | 0.25 |
| CD3 e/d | 0.09 | 0.15 | 0.60 |
| CD15 epi | 5050 | 1871 | 2.70 |
| CD15 dermis | 743 | 387 | 1.92 |
| CD15 e/d | 6.80 | 4.83 | 1.41 |
| CD123 epi | 1038 | 801 | 1.30 |
| CD123 dermis | 2084 | 1285 | 1.62 |
| CD123 e/d | 0.50 | 0.62 | 0.81 |
| CD11c epi | 6174 | 3237 | 1.91 |
| CD11c dermis | 4643 | 4257 | 1.09 |
| CD11c e/d | 1.33 | 0.76 | 1.75 |

Table 3: All numbers are for the lesion except the second row. The cell counts are per mm2. The widths and seperation are measured in um.

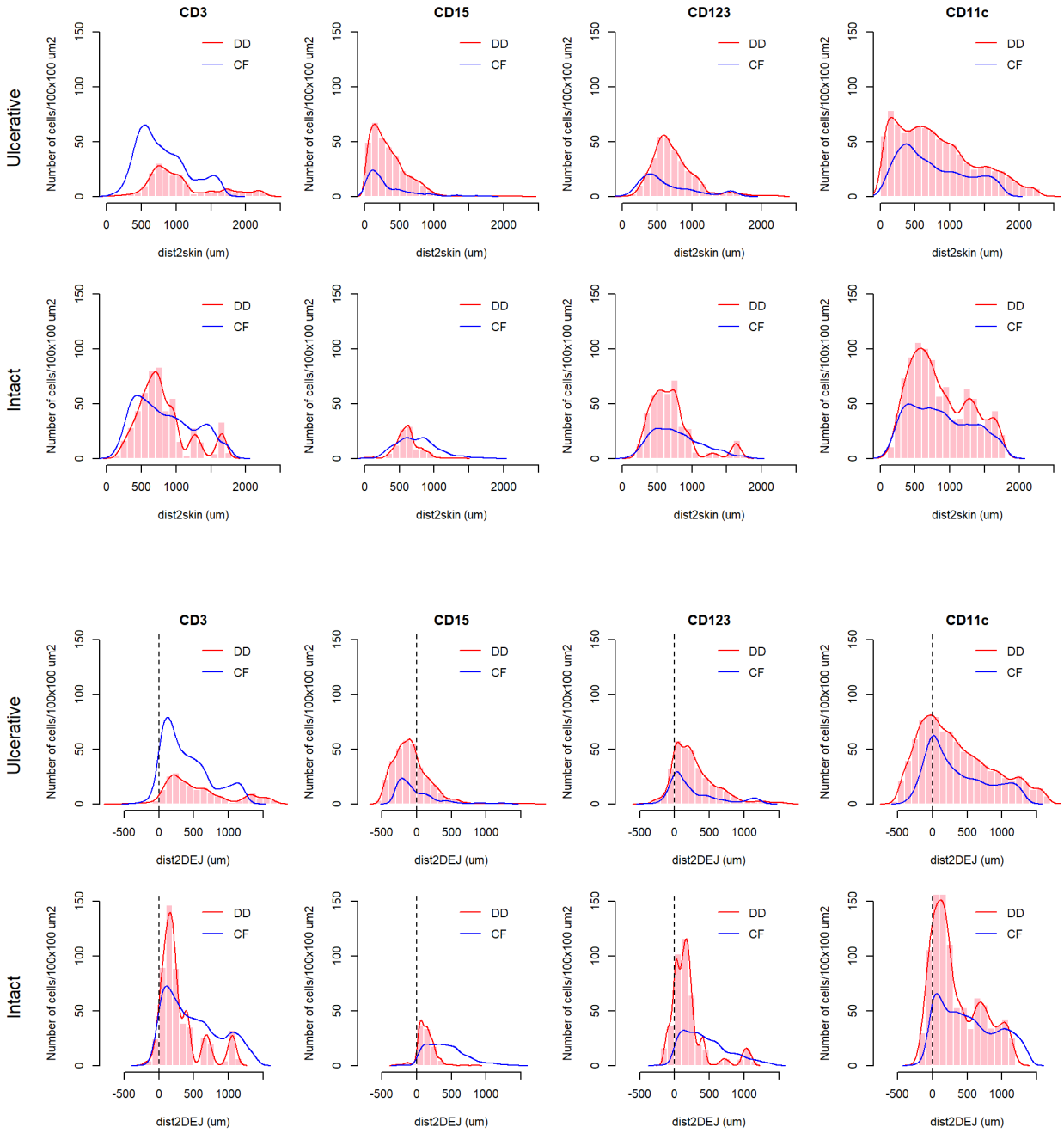



Figure 1: Number of cells per 100x100 um2 tissue as a function of distance to DEJ or skin. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width. For example, CD123 epi count 1038/2862*100=36.

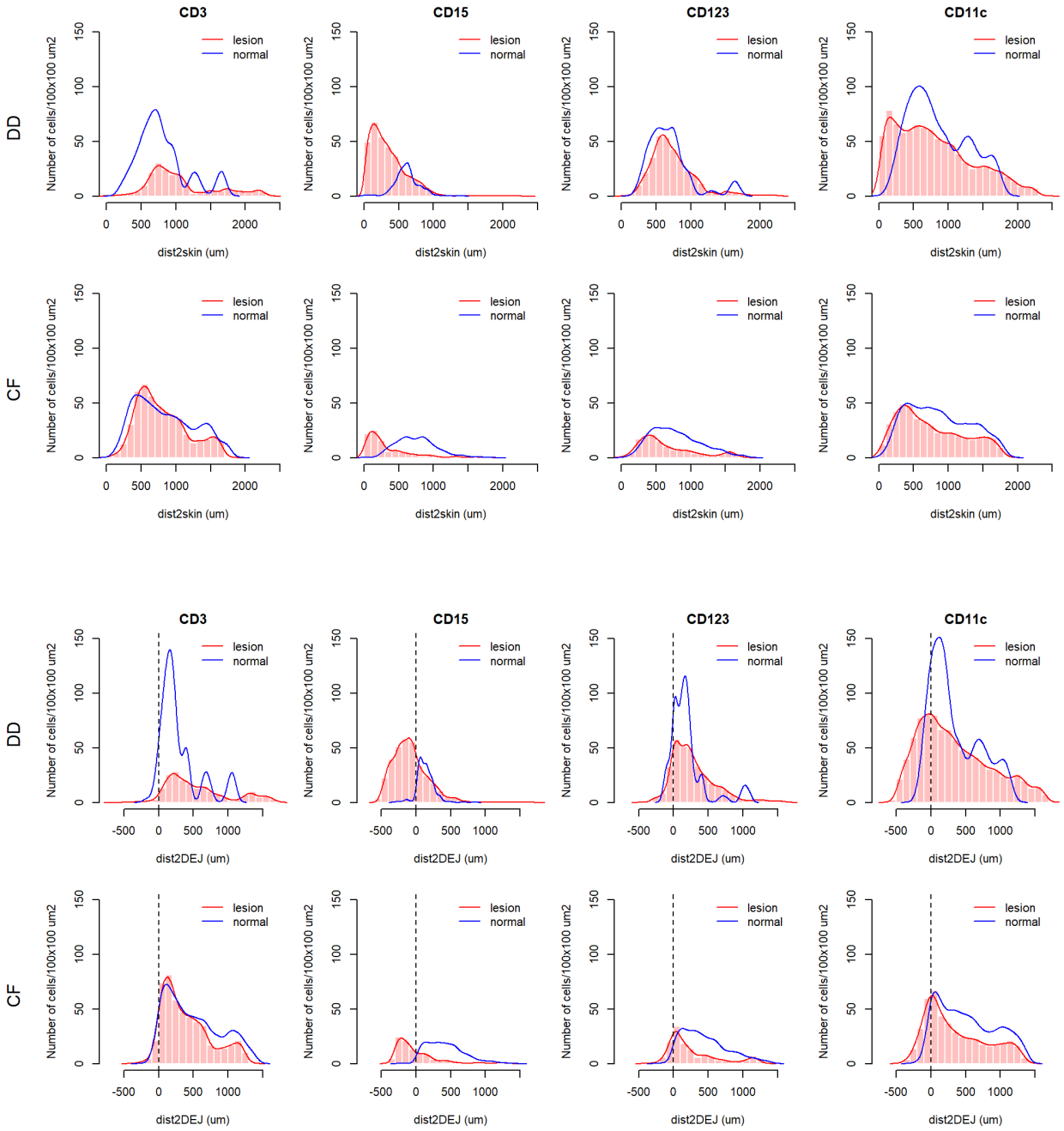



Figure 2: Number of cells per 100x100 um2 tissue as a function of distance to skin or DEJ. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width. For example, CD123 epi count 1038/2862*100=36.

The ratio between lesion and normal is about 1 for CD3/CF, but about 1/3 in CD3/DD. This suggests that CD3 is depleted in lesion in DD, but not in CF.

Compare lesion with normal in CD15/CF, we see that the lesion peak is shifted to the left compared to the normal peak. This pattern is more clear in CF than in DD. We don't see this pattern for CD3.

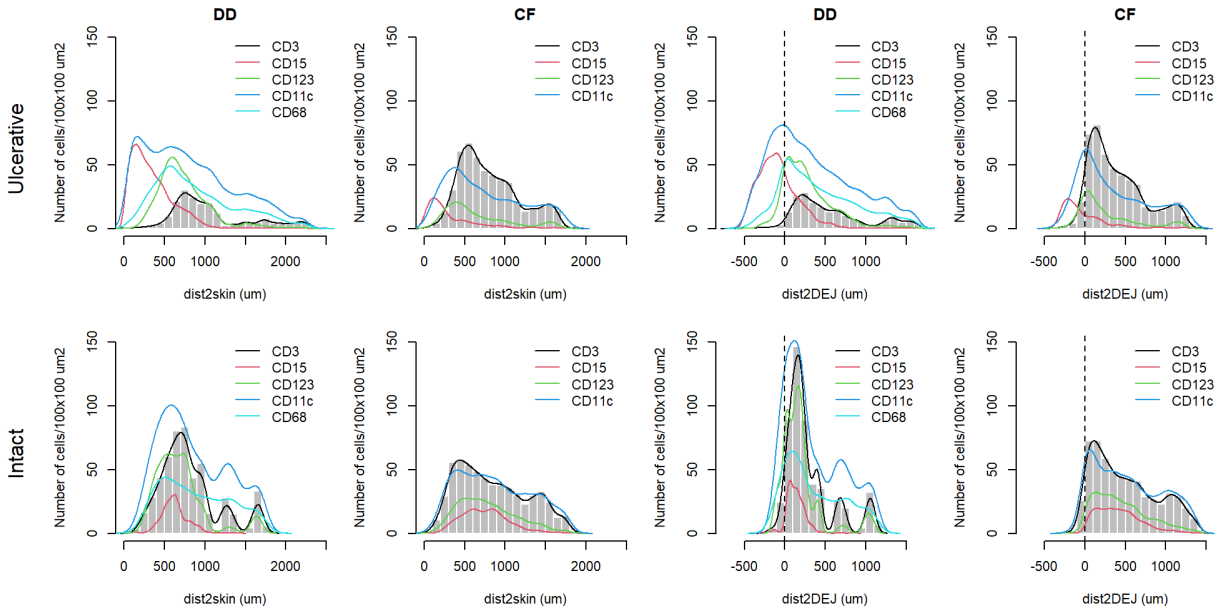



Figure 3: Number of cells per 100x100 um2 tissue as a function of distance to skin. The curves are smoothed over histograms and each bin in the histogram represents a 100 um interval. Adding up the counts from all the bins gives the number of cells in tissue of 100 um in width.

That the CD15 dist2skin peak in DD/lesion is near skin while the corresponding dist2DEJ peak is near DEJ is due to the fact that the DEJ line is very curvy.

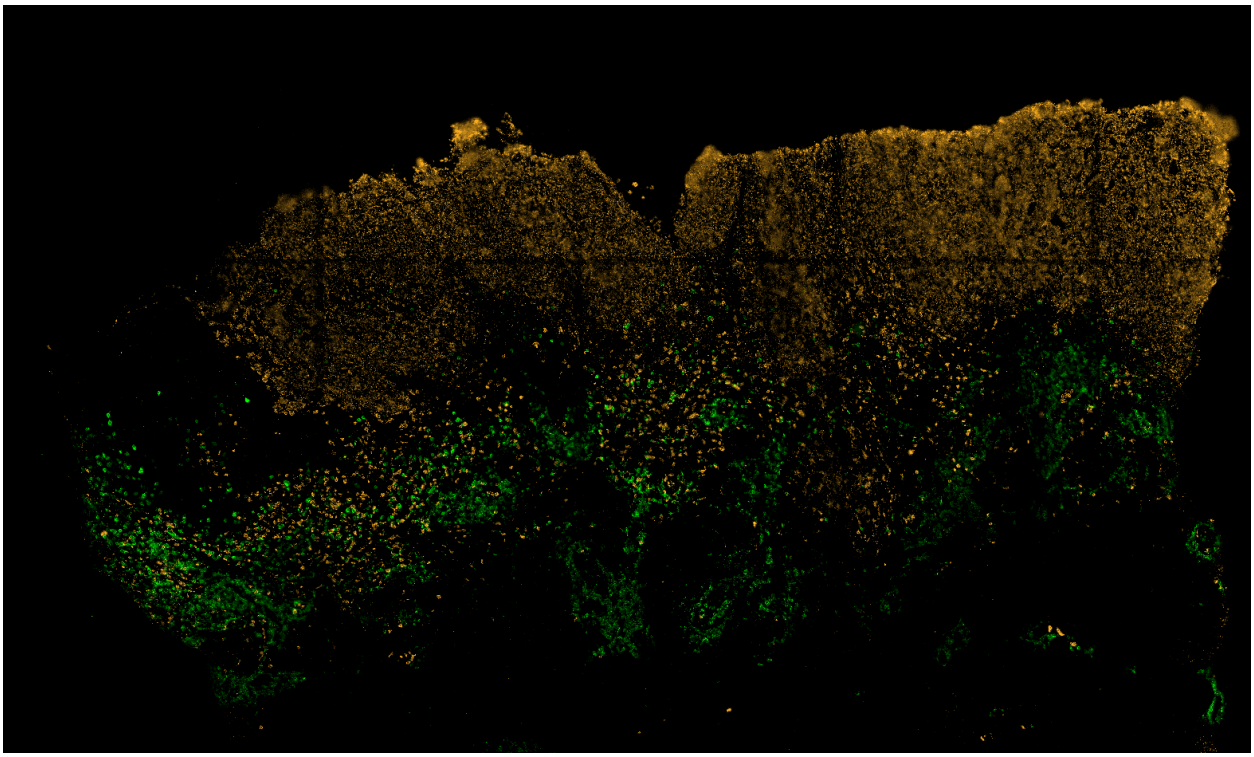


Figure 4: DD, CD15 gold, CD3 green.

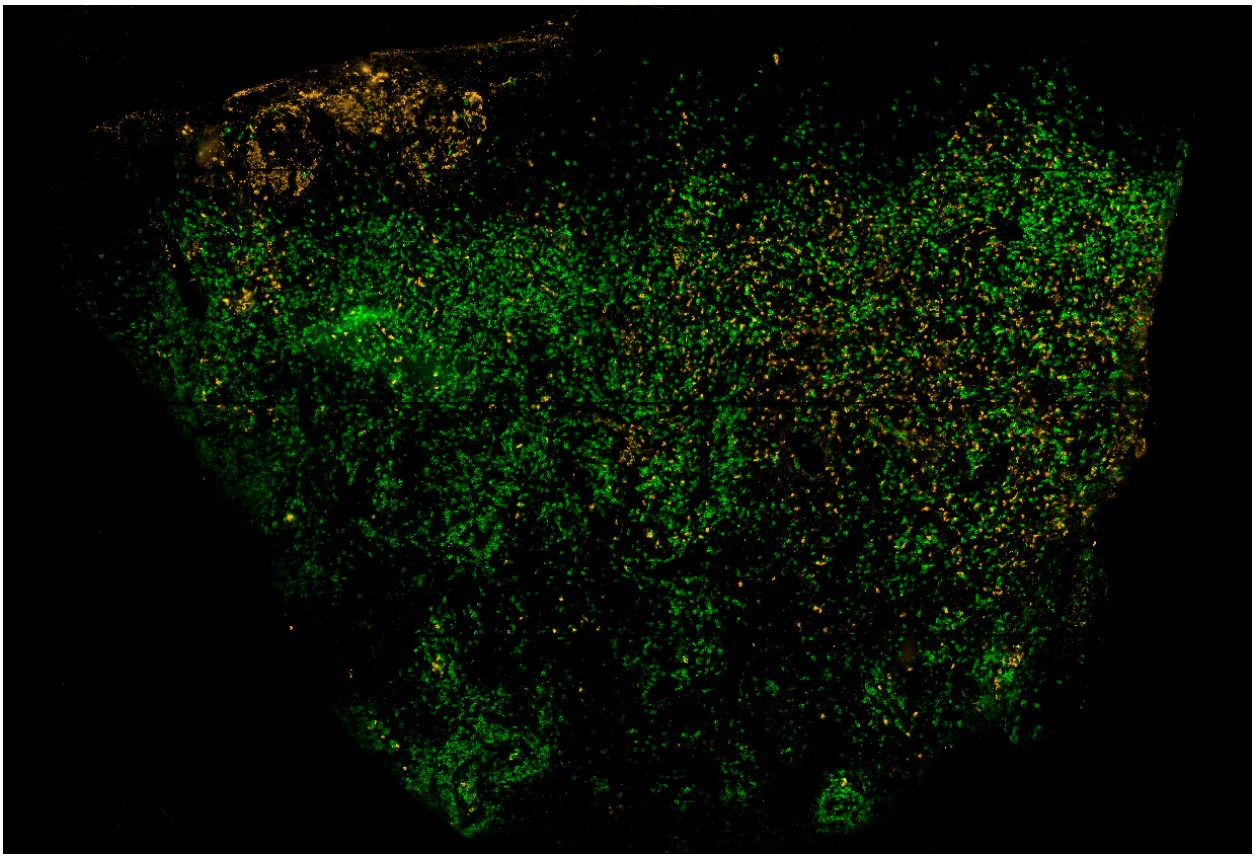


Figure 5: CF, CD15 gold, CD3 green.